

Devops Assignment 1

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Task 1.1 -

```
PS C:\WINDOWS\System32> cd C:\Users\dhruv\Downloads
PS C:\Users\dhruv\Downloads> mkdir hello-world-flask

Directory: C:\Users\dhruv\Downloads

Mode                LastWriteTime         Length Name
----                -
d-----          08-07-2026         13:28         hello-world-flask

PS C:\Users\dhruv\Downloads> cd hello-world-flask
PS C:\Users\dhruv\Downloads\hello-world-flask> git init
Initialized empty Git repository in C:/Users/dhruv/Downloads/hello-world-flask/.git/
PS C:\Users\dhruv\Downloads\hello-world-flask> git branch -M main
PS C:\Users\dhruv\Downloads\hello-world-flask> code .
PS C:\Users\dhruv\Downloads\hello-world-flask> git add .
PS C:\Users\dhruv\Downloads\hello-world-flask> git commit -m "Initial commit: Hello World Flask app"
[main (root-commit) 73afd3b] Initial commit: Hello World Flask app
 2 files changed, 11 insertions(+)
 create mode 100644 app.py
 create mode 100644 requirements.txt
PS C:\Users\dhruv\Downloads\hello-world-flask> git log --oneline
73afd3b (HEAD -> main) Initial commit: Hello World Flask app
PS C:\Users\dhruv\Downloads\hello-world-flask> git status
On branch main
nothing to commit, working tree clean
```

Task 1.2 –

```

origin https://github.com/DhurvG/hello-world-flask.git (push)
PS C:\Users\dhruv\Downloads\hello-world-flask> git push -u origin main
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 8 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (4/4), 422 bytes | 84.00 KiB/s, done.
Total 4 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/DhurvG/Hello-World-Flask.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.
PS C:\Users\dhruv\Downloads\hello-world-flask> git push -u origin main
branch 'main' set up to track 'origin/main'.
Everything up-to-date
PS C:\Users\dhruv\Downloads\hello-world-flask> git push -u origin feature/user-auth
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 8 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 428 bytes | 214.00 KiB/s, done.
Total 3 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
remote:
remote: Create a pull request for 'feature/user-auth' on GitHub by visiting:
remote:   https://github.com/DhurvG/Hello-World-Flask/pull/new/feature/user-auth
remote:
To https://github.com/DhurvG/Hello-World-Flask.git
 * [new branch]      feature/user-auth -> feature/user-auth
branch 'feature/user-auth' set up to track 'origin/feature/user-auth'.

```

Task 1.3 –

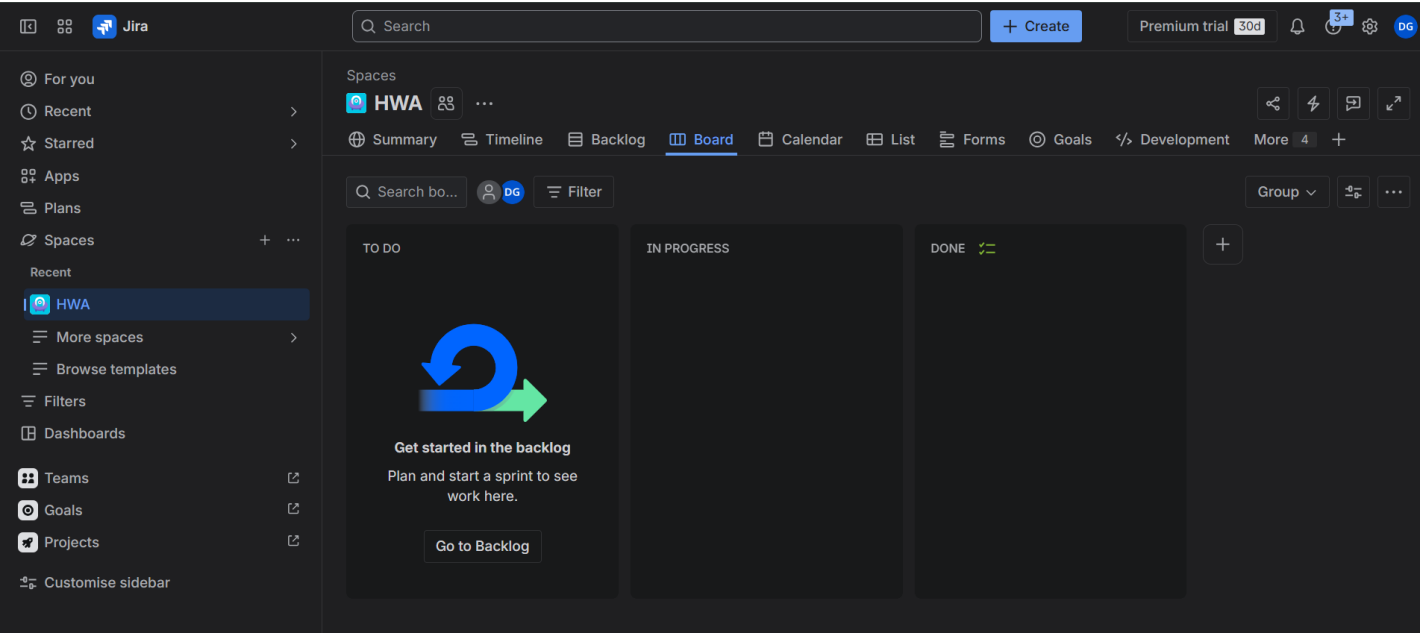
```

PS C:\Users\dhruv\Downloads\hello-world-flask> git add app.py
PS C:\Users\dhruv\Downloads\hello-world-flask> git commit -m "Update print statement on main branch"
[main 6bd5dde] Update print statement on main branch
 1 file changed, 6 insertions(+), 1 deletion(-)
PS C:\Users\dhruv\Downloads\hello-world-flask> git merge feature/user-auth
Auto-merging app.py
CONFLICT (content): Merge conflict in app.py
Automatic merge failed; fix conflicts and then commit the result.
PS C:\Users\dhruv\Downloads\hello-world-flask> git add app.py
PS C:\Users\dhruv\Downloads\hello-world-flask> git commit -m "Resolve merge conflict between main and feature/user-auth"

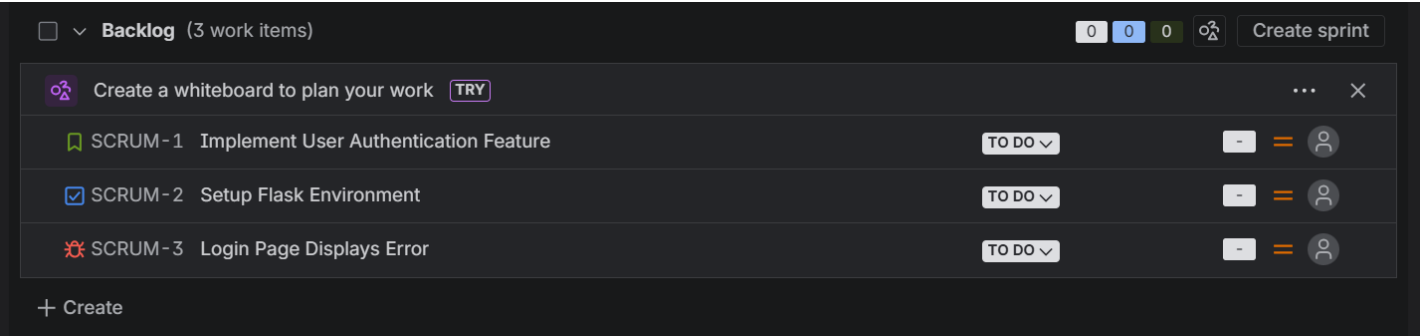
[main c033261] Resolve merge conflict between main and feature/user-auth
PS C:\Users\dhruv\Downloads\hello-world-flask> git push origin main
Enumerating objects: 10, done.
Counting objects: 100% (10/10), done.
Delta compression using up to 8 threads
Compressing objects: 100% (6/6), done.
Writing objects: 100% (6/6), 716 bytes | 179.00 KiB/s, done.
Total 6 (delta 2), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (2/2), completed with 1 local object.
To https://github.com/DhurvG/Hello-World-Flask.git
 73afd3b..c033261  main -> main
PS C:\Users\dhruv\Downloads\hello-world-flask> git log --graph --oneline --all
*   c033261 (HEAD -> main, origin/main) Resolve merge conflict between main and feature/user-auth
|
| * c649497 (origin/feature/user-auth, feature/user-auth) Add placeholder print statement for user-auth branch
| * 6bd5dde Update print statement on main branch
|/
* 73afd3b Initial commit: Hello World Flask app
PS C:\Users\dhruv\Downloads\hello-world-flask>

```

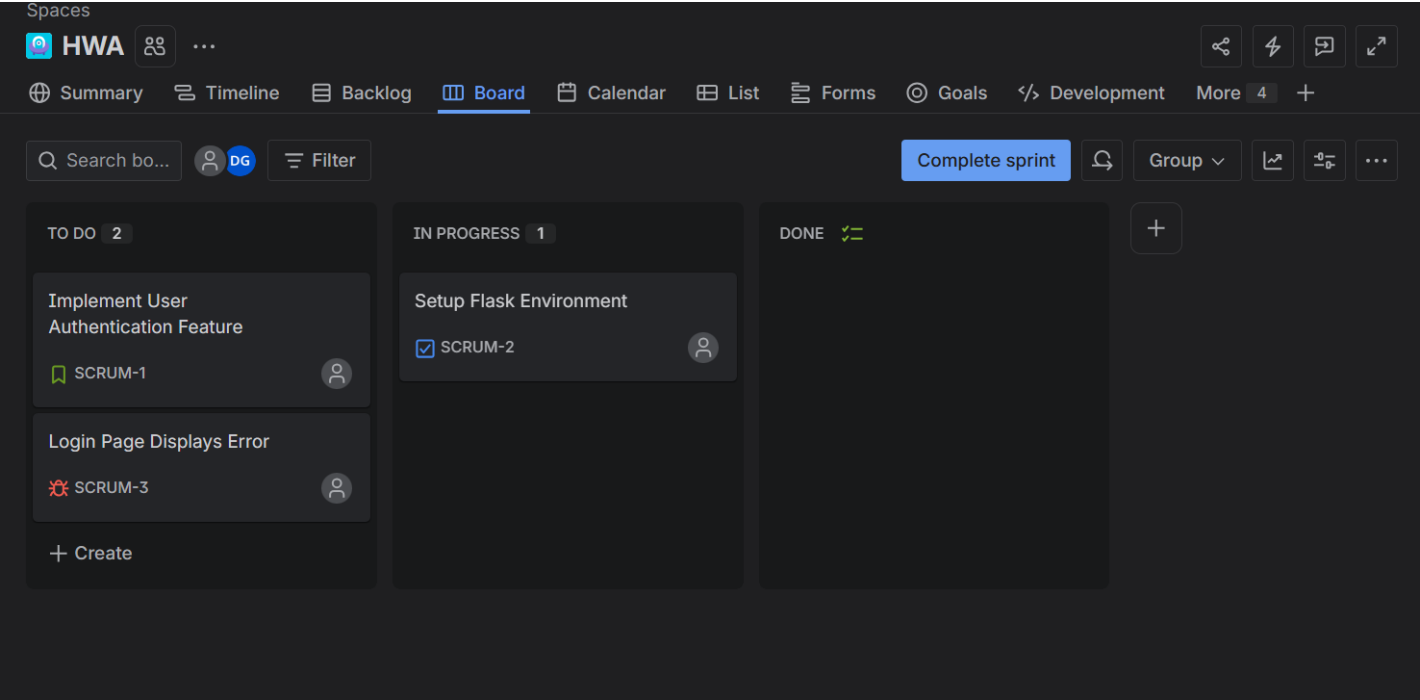
Task 2.1-



Task 2.2 –



Task 2.3 –



Task 3.1 –

```
PS C:\Users\dhruv\Downloads\hello-world-flask> docker build -t hello-world-flask .
[+] Building 29.9s (11/11) FINISHED                                docker:desktop-linux
=> [internal] load build definition from Dockerfile                0.2s
=> => transferring dockerfile: 215B                                0.0s
=> [internal] load metadata for docker.io/library/python:3.11-slim 4.5s
=> [auth] library/python:pull token for registry-1.docker.io       0.0s
=> [internal] load .dockerignore                                   0.0s
=> => transferring context: 2B                                     0.0s
=> [1/5] FROM docker.io/library/python:3.11-slim@sha256:e031123e3d85762b141ad1cbc56452ba69c6e722ebf2f042cc0dc86 15.1s
=> => resolve docker.io/library/python:3.11-slim@sha256:e031123e3d85762b141ad1cbc56452ba69c6e722ebf2f042cc0dc86c 0.1s
=> => sha256:3ec520ed9418633eed929594af6973fdacd61a7f74223b9b9227ee0a8effe2ea 14.42MB / 14.42MB 4.1s
=> => sha256:f66cea3b82f91b150204030cc52f015b3432cabcf6857125c94da9006f76b55b 250B / 250B 1.3s
=> => sha256:298bacc1f3a58135e8bf2182d29ba0c456b07c2a04284aa2542fa5855374118d 1.29MB / 1.29MB 3.7s
=> => sha256:e95a6c7ea7d49b37920899b023ecd0e32796c976c1748491f76cae53ba86d13a 29.79MB / 29.79MB 11.9s
=> => extracting sha256:e95a6c7ea7d49b37920899b023ecd0e32796c976c1748491f76cae53ba86d13a 1.7s
=> => extracting sha256:298bacc1f3a58135e8bf2182d29ba0c456b07c2a04284aa2542fa5855374118d 0.2s
=> => extracting sha256:3ec520ed9418633eed929594af6973fdacd61a7f74223b9b9227ee0a8effe2ea 0.9s
=> => extracting sha256:f66cea3b82f91b150204030cc52f015b3432cabcf6857125c94da9006f76b55b 0.0s
=> [internal] load build context                                  0.2s
=> => transferring context: 473B                                    0.1s
=> [2/5] WORKDIR /app                                           0.3s
=> [3/5] COPY requirements.txt .                                  0.1s
=> [4/5] RUN pip install --no-cache-dir -r requirements.txt      7.7s
=> [5/5] COPY app.py .                                           0.1s
=> exporting to image                                           1.5s
=> => exporting layers                                             0.9s
=> => exporting manifest sha256:9680458a4ad48f2977aace6f0645f3592ccc85f415da372675d3036fe7dc8c94 0.0s
=> => exporting config sha256:b6ef1955b5a98c933aa67faf4a5919b4e3774b2826a6b86463cce07b4c1a664a 0.0s
=> => exporting attestation manifest sha256:4286396156461f0a849d2ab2cf78d16091bf2295a3bdc5462829396e77723bdb 0.0s
=> => exporting manifest list sha256:26596d97a3e712707172af92beb3365307c61f1404af0eb033c46ccaa0b25dc4 0.0s

PS C:\Users\dhruv\Downloads\hello-world-flask> docker run -d -p 5000:5000 --name hwf-container hello-world-flask
d42d3f08d1317f8a67345494b8c63c01548aa7669e7879e43d10c927f3ef4505
PS C:\Users\dhruv\Downloads\hello-world-flask> docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS
NAMES
d42d3f08d131  hello-world-flask  "python app.py"         24 seconds ago Up 23 seconds  0.0.0.0:5000->5000/tcp, [::]:5000->5000/tcp  hwf-container
PS C:\Users\dhruv\Downloads\hello-world-flask> curl http://localhost:5000

Security Warning: Script Execution Risk
Invoke-WebRequest parses the content of the web page. Script code in the web page might be run when the page is
parsed.
RECOMMENDED ACTION:
Use the -UseBasicParsing switch to avoid script code execution.

Do you want to continue?

[Y] Yes [A] Yes to All [N] No [L] No to All [S] Suspend [?] Help (default is "N"): A

StatusCode      : 200
StatusDescription : OK
Content          : Hello, World!
RawContent       : HTTP/1.1 200 OK
                  Connection: close
                  Content-Length: 13
                  Content-Type: text/html; charset=utf-8
                  Date: Wed, 08 Jul 2026 17:39:04 GMT
                  Server: Werkzeug/3.1.8 Python/3.11.15

                  Hello, World!
Forms           : {}
Headers         : {[Connection, close], [Content-Length, 13], [Content-Type, text/html; charset=utf-8], [Date, Wed,
                  08 Jul 2026 17:39:04 GMT]...}
Images          : {}
InputFields     : {}
Links           : {}
ParsedHtml      : mshtml.HTMLDocumentClass
RawContentLength : 13
```

Task 3.2-

Console

```
Started by user Dhruv Gangurde
Running as SYSTEM
Building in workspace /var/jenkins_home/workspace/hello-world-flask-build
The recommended git tool is: NONE
No credentials specified
Cloning the remote Git repository
Cloning repository https://github.com/DhurvG/Hello-World-Flask.git
> git init /var/jenkins_home/workspace/hello-world-flask-build # timeout=10
Fetching upstream changes from https://github.com/DhurvG/Hello-World-Flask.git
> git --version # timeout=10
> git --version # 'git version 2.47.3'
> git fetch --tags --force --progress -- https://github.com/DhurvG/Hello-World-Flask.git +refs/heads/*:refs/remotes/origin/* # timeout=10
> git config remote.origin.url https://github.com/DhurvG/Hello-World-Flask.git # timeout=10
> git config --add remote.origin.fetch +refs/heads/*:refs/remotes/origin/* # timeout=10
Avoid second fetch
> git rev-parse refs/remotes/origin/main^{commit} # timeout=10
Checking out Revision fc5c99c7899d36d0eb0d334f64f627c78b0c0f81 (refs/remotes/origin/main)
> git config core.sparsecheckout # timeout=10
> git checkout -f fc5c99c7899d36d0eb0d334f64f627c78b0c0f81 # timeout=10
Commit message: "Add Dockerfile for containerization"
First time build. Skipping changelog.
[hello-world-flask-build] $ /bin/sh -xe /tmp/jenkins1325692542953656993.sh
+ ls -la
total 24
drwxr-xr-x 3 jenkins jenkins 4096 Jul  9 13:13 .
drwxr-xr-x 3 jenkins jenkins 4096 Jul  9 13:12 ..
drwxr-xr-x 8 jenkins jenkins 4096 Jul  9 13:13 .git
-rw-r--r-- 1 jenkins jenkins 165 Jul  9 13:13 Dockerfile
-rw-r--r-- 1 jenkins jenkins 373 Jul  9 13:13 app.py
-rw-r--r-- 1 jenkins jenkins  5 Jul  9 13:13 requirements.txt
Finished: SUCCESS
```